Adityaa Saha

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Technical Skills

Languages: C, C++, JavaScript, TypeScript, Python **Web:** React, Node.js, Express
Databases: MySQL, MongoDB **Tools:** Git, GitHub **Concepts:** DSA, OOP, COA, DBMS, CN, OS

Projects

32-bit RISC-V CPU Simulator C/C++

- Implemented a 32-bit pipelined RV32I processor simulator with a 5-stage pipeline (IF, ID, EX, MEM, WB).
- Added hazard detection, data forwarding, and pipeline stall simulation.

GitHub Repository

LSM Tree Key-Value Store C++

- Built a disk-based KV store using LSM Tree with MemTable, SSTables, and leveled compaction.
- Integrated WAL for crash recovery and CRC32 checksums for data integrity.

GitHub Repository

DNS Server Implementation C++

- Built a multithreaded DNS server over UDP and TCP with a thread pool, authoritative zone files, and upstream query forwarding.
- Added response caching, per-client rate limiting, domain blocklists, rotating logs, and an HTTP /metrics endpoint.

GitHub Repository

HTTP Load Balancer C++

- Built an HTTP load balancer in C++17 with a custom socket layer, HTTP parser, and pluggable backend scheduling.
- Implemented active health checks, connection management, runtime configuration, and live metrics.

GitHub Repository

Education

Indian Institute of Technology Bhubaneswar 2023 – 2027
B.Tech in Computer Science and Engineering

Positions of Responsibility

- Core Member, Dramatics Society — The Fourth Wall
- Event Management Team Member, Cultural Fest — Alma Fiesta

Extracurricular Activities

- Member of the IIT Bhubaneswar football team; active in dramatics and stage performances

Achievements

- Secured 2nd Place in Inter-Branch General Championship (Game Development)

Interests

Data Structures and Algorithms, Systems Programming, Backend Development